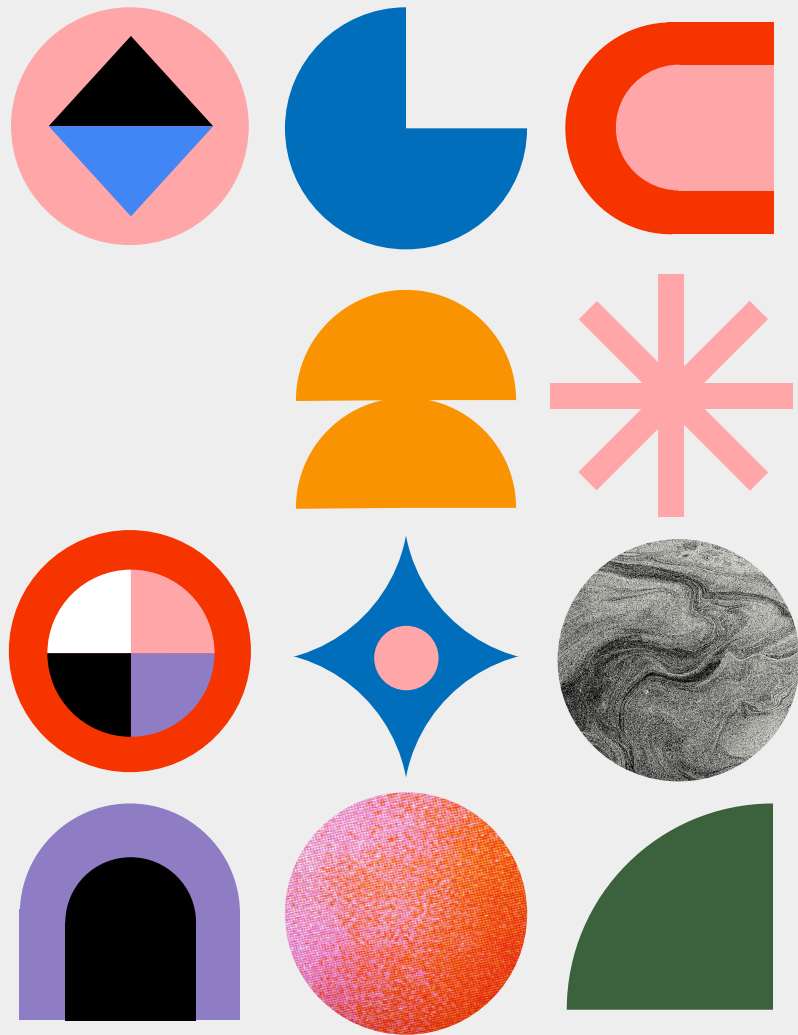


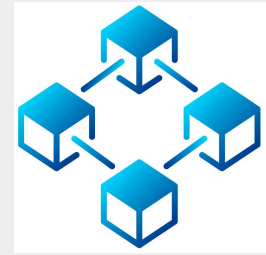
Lab - 6

Hyperledger Fabric

Prepared by
Sajid Imam Mahir (SJM)



Public vs Private Blockchain



Public	Private/ Permissioned/ Consortium
Anyone can join pseudonymously	Participation restricted according to org
Cannot pinpoint a specific node	Every node is uniquely identifiable
Requires high gas-fee	No gas fee required
All transactions are public	Transactions are restricted inside org
Slow transaction approval	Almost instant transaction approval
Ex: Bitcoin, Ethereum, Solana etc	Ex: Hyperledger Fabric, Corda etc



Why Fabric?



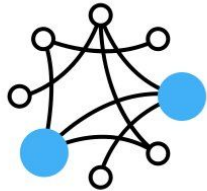
Hyperledger Fabric is an **open source enterprise-grade permissioned distributed ledger technology** (DLT) platform

- **Known Identities:** Digital IDs issued by a **Certificate Authority**
- **Free to Transact:** There is no "Fabric coin"
- **Privacy:** Offers private communication **channel**
- **Control:** Multiple **organizations** maintain and control it
- **Smart Contracts:** Programming logic executed using "**Chaincode**."
- **Speed:** Fabric can process around **3500 tx/s**. Compare this with only **3-5 tx/s** for **Bitcoin** and around **10 tx/s** for **Ethereum**.

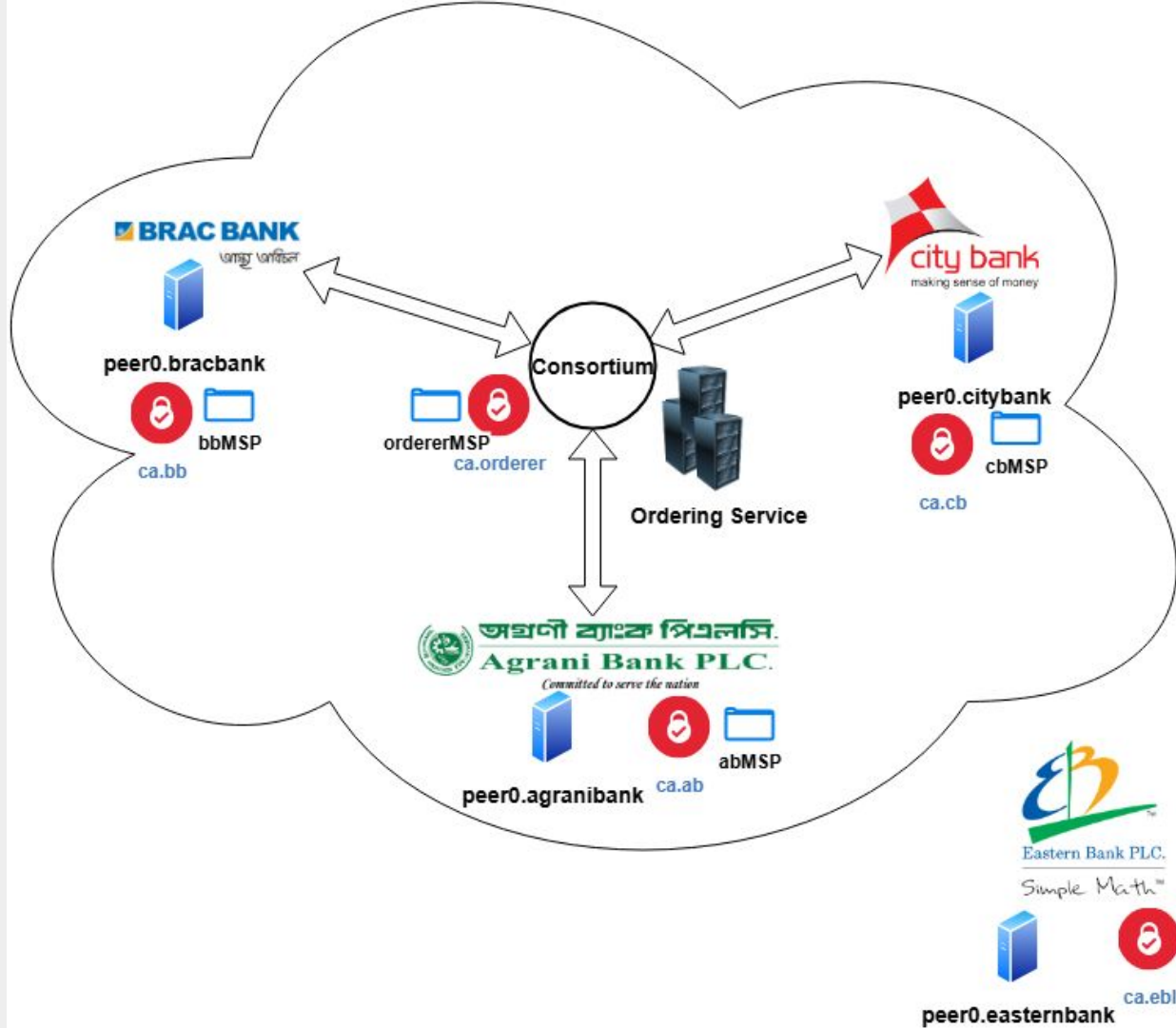


Consortium

A group of organizations with a **shared destiny** or business goal that **agree to govern a blockchain network** together

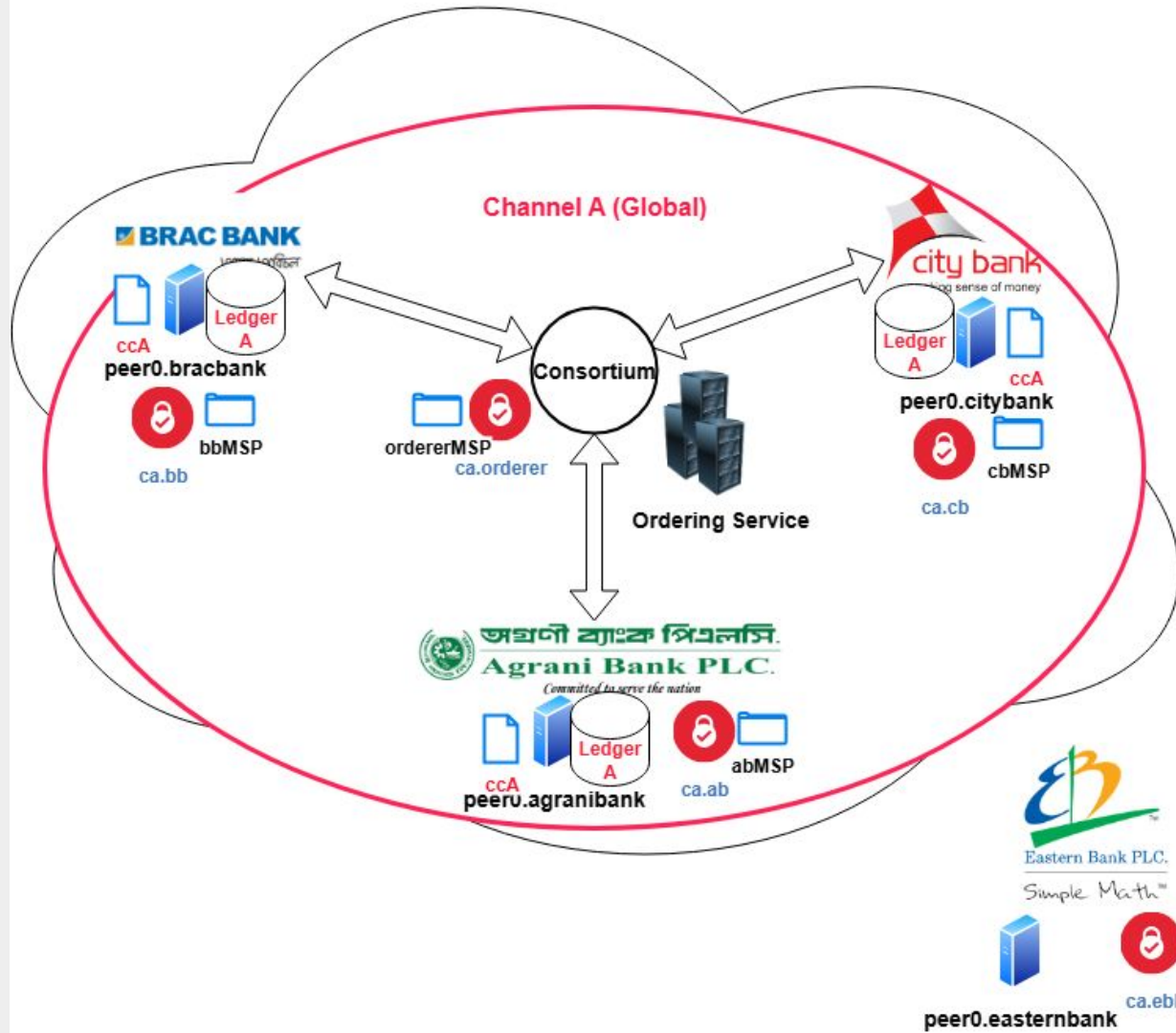


Consortium Blockchain



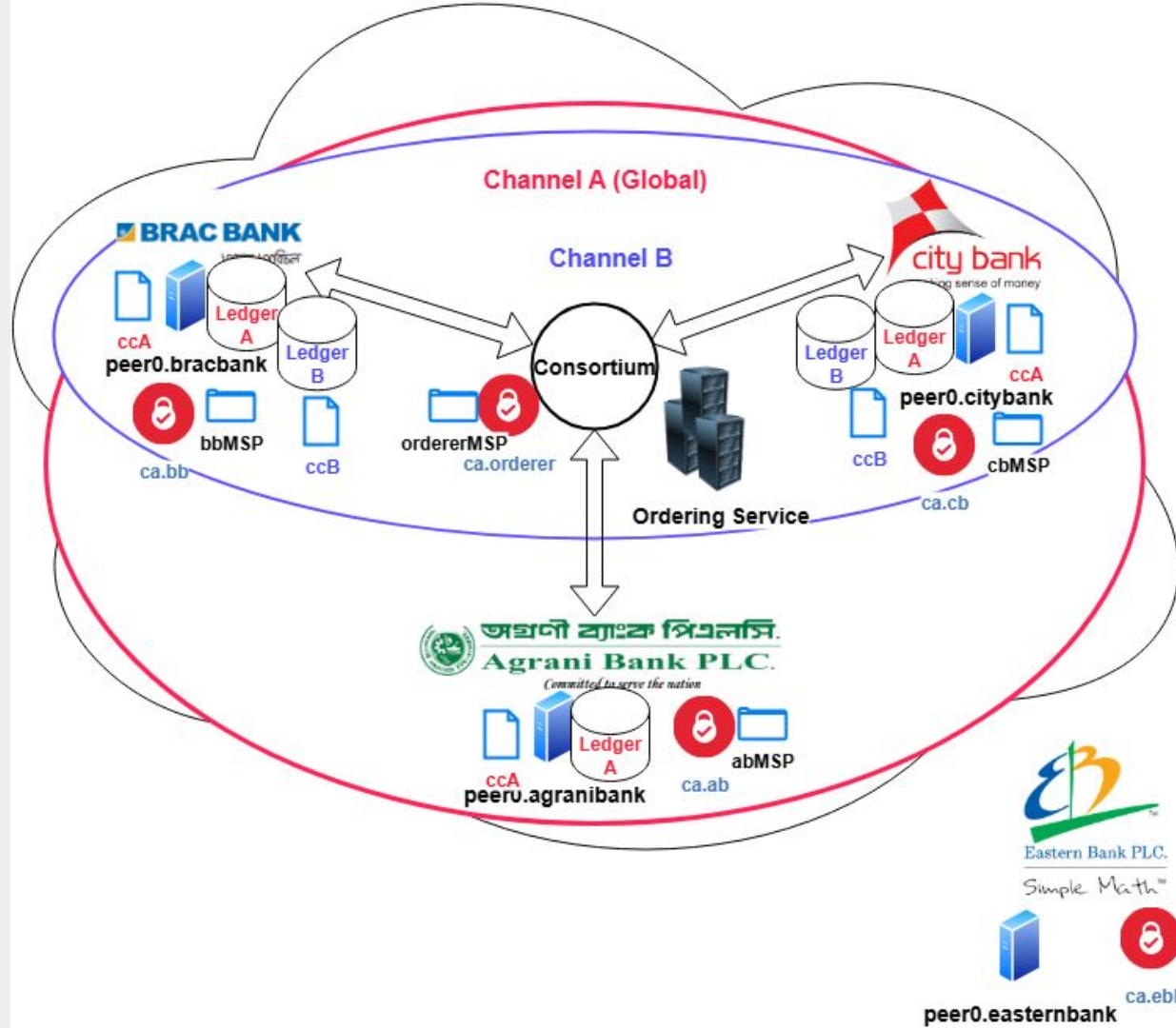
Single Channel

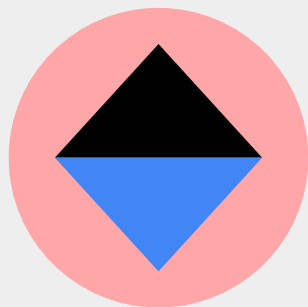
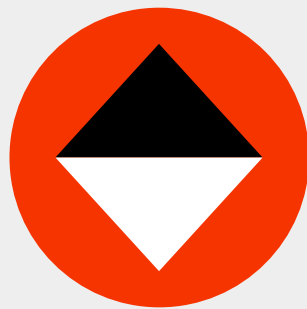
A private "subnet" or dedicated communication mechanism between a specific subset of organizations.



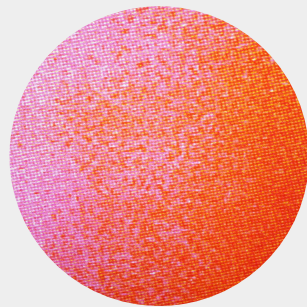
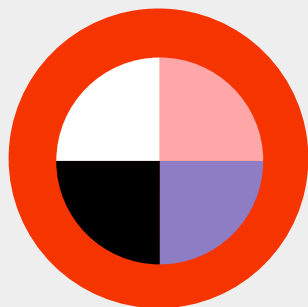
Multi Channel

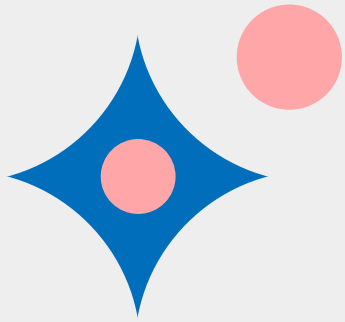
Each peer **only shares** the ledger with nodes that are in the **same channel**





PART 2





Fabric Nodes & Different Roles

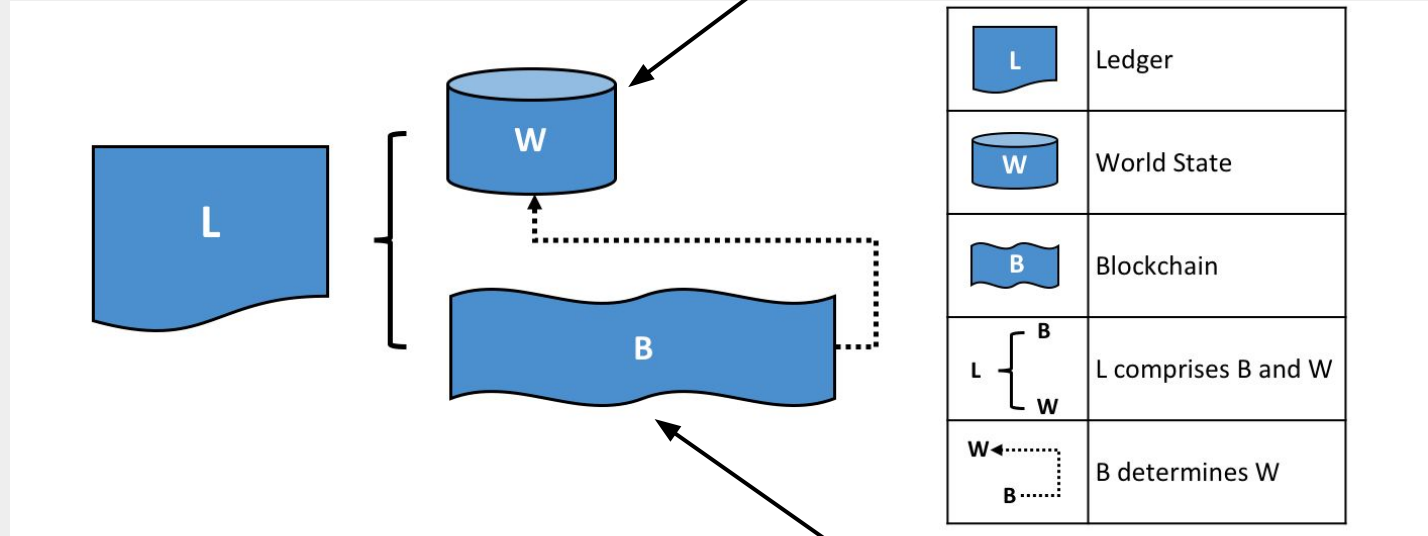
- **Certificate Authority (CA)** : Certificate Issuer
- **Membership Service Provider (MSP)** : Role verifying list

- **Committing Peer** : Validator
- **Endorsing Peer** : Executor + Validator
- **Ordering Service Node** : Organizer

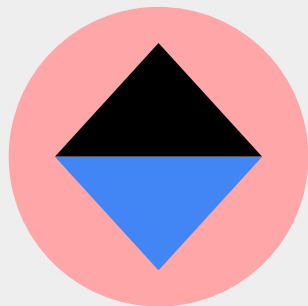
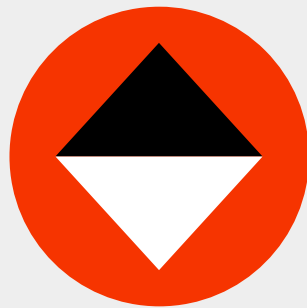
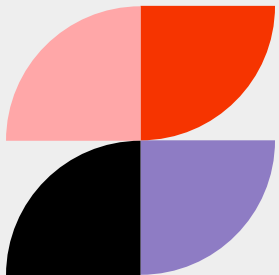


The Ledger

The current State
of the DB



Transaction log of
all records



Thank you

